

**Department of Computer Science & Engineering (IoT)****Vision of the Department***To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.***Mission of the Department***To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skillsthrough emerging technologies.***Session 2026-2027****Vision:** Dream of where you want.**Mission:** Means to achieve Vision**Program Educational Objectives of the program (PEO):** (broad statements that describe the professional and career accomplishments)

PEO1	Preparation	P: Preparation	Pep-CL abbreviation pronounce as Pep-si-IL easy to recall
PEO2	Core Competence	E: Environment (Learning Environment)	
PEO3	Breadth	P: Professionalism	
PEO4	Professionalism	C: Core Competence	
PEO5	Learning Environment	L: Breadth (Learning in diverse areas)	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)**Keywords of POs:**

Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

“I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life.” to contribute to the development of cutting-edge technologies and Research.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.**Sheikh Wasimuddin**
Name of Student

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Session	2026-27 (ODD)	Course Name	Lab : Embedded Systems
Semester	7	Course Code	23IOT1706
Roll No	61	Name of Student	Sheikh Wasimuddin

Practical Number	03
Course Outcome	CO1: Explain the fundamentals of embedded microcontrollers, development environments, and simulation tools. CO2: Develop embedded applications using Arduino/ESP32 and interface sensors, actuators, and peripherals. CO3: Implement event-driven and communication-based embedded systems using interrupts, UART, and I2C protocols. CO4: Design, simulate, and test embedded system applications for real-world applications.
Aim	To interface a servo motor with Arduino Uno and rotate it to different angles (0°, 90°, and 180°).
Theory (100 words)	A servo motor is an electromechanical device used to control the angular position of a mechanical component with precision. Unlike a conventional DC motor, which continuously rotates when power is supplied, a servo motor can be commanded to move to a specific angle and hold that position. A typical servo motor has three connections: VCC, GND, and Signal. VCC supplies power, GND provides the common reference, and the Signal pin receives control pulses from the microcontroller. An Arduino Uno can control a servo using the Servo library and the write() function. In this practical, the servo is connected to digital pin 9 of the Arduino Uno. The program commands the motor to move successively to 0°, 90°, and 180° with suitable time delays. This demonstrates the use of Arduino for precise angular control and servo interfacing. Servo motors are commonly used in robotic arms, automation systems, camera positioning, and model-control applications.

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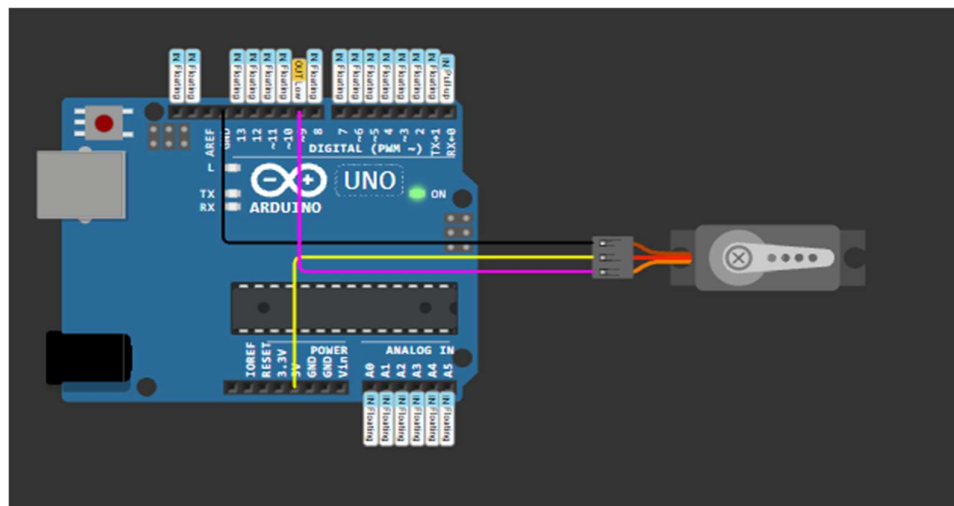
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Procedure
and
Execution

(100 Words)

1. Open Tinkercad and select Circuits.
2. Create a new circuit and place an Arduino Uno R3 on the workspace.
3. Add a Servo Motor component to the circuit.
4. Connect the VCC wire of the servo to the Arduino 5V pin.
5. Connect the GND wire of the servo to the Arduino GND pin.
6. Connect the Signal wire of the servo to Digital Pin 9 of the Arduino.
7. Open the Code section and select Text mode.
8. Write the Arduino program using the Servo library.
9. Compile the program and check that there are no errors.
10. Start the Tinkercad simulation.
11. Observe the servo rotating to 0° and holding its position.
12. Observe the servo rotating to 90° after the specified delay.
13. Observe the servo rotating to 180° after another delay.
14. Verify that the servo repeats the sequence 0° → 90° → 180° continuously.
15. Record the circuit, program, output, and observations as screenshots in the practical file.
16. Stop the simulation after verifying the correct operation.

Stepwise Screenshots with steps:





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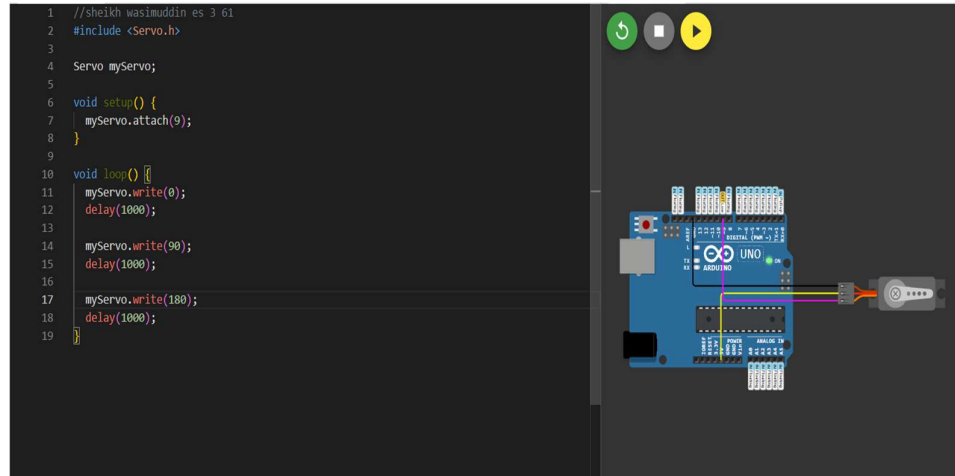
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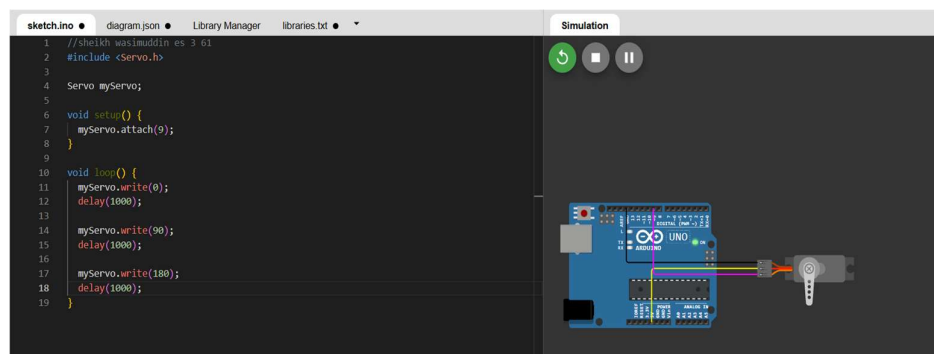
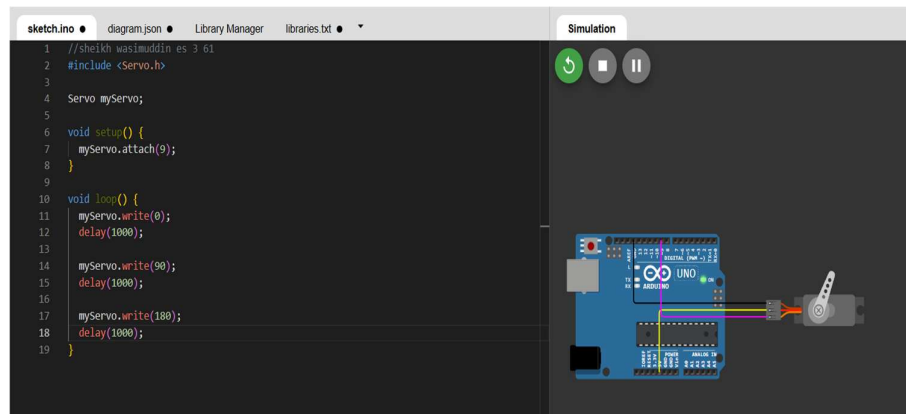
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Conclusion	The servo motor was successfully interfaced with the Arduino Uno and controlled using the Servo library . The motor was programmed to rotate to 0°, 90°, and 180° at specified time intervals. The practical demonstrated how Arduino can be used to control the precise angular position of a servo motor. It also provided an understanding of servo interfacing, signal control, and PWM-based operation , which are useful in robotics, automation, and embedded systems applications.
Date	20-08-2026